Year 12 Maths - Statistical hypothesis testing

Similar questions - Sampling and random selection

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A school has 800 students: 360 in Years 7–9, 240 in Years 10–11 and 200 in Years 12–13. The head wants a sample of 40 students to answer a survey about the canteen.

(a) Describe how to take a sample of 40 that is stratified by year group, and calculate how many students come from each group. **[3]**

(b) Give one advantage of a stratified sample over a simple random sample in this context. **[1]**

(c) The head suggests instead asking the first 40 students who arrive at the canteen at morning break. Give two reasons why this sample will not be random. **[2]**

(d) Describe how a systematic sample of size 40 could be taken from an alphabetical list of the 800 students. **[2]**

2 A teacher has 12 students numbered 1 to 12 and wants to choose one of them at random using a fair six-sided die. She suggests throwing two dice and choosing the student whose number is the total score on the two dice (the total ranges from 2 to 12, so students 1 is never chosen).

(a) Show that this method is not random. **[2]**

(b) Describe a method, using a single die thrown twice, that gives each of the 12 students the same probability of being chosen, and show that this probability is $\frac{1}{12}$. **[3]**

Question 1

(a) Sample size from each group $= \dfrac{\text{group size}}{800} \times 40$: Years 7–9: 18; Years 10–11: 12; Years 12–13: 10. Choose each group's students by simple random sampling

(b) The sample is guaranteed to contain the right proportion from every year group, so each year is represented (a simple random sample might by chance have few or no sixth-form students)

(c) Students who arrive first may be those who are first out of lessons or who eat early (so not representative of all years); also not every student has an equal chance of being selected (students who stay in class or arrive late cannot be chosen)

(d) Sampling interval $\dfrac{800}{40} = 20$. Choose a random starting number between 1 and 20 and then take every 20th student on the list

Question 2

(a) The totals are not equally likely: a total of 2 can happen in 1 way $\left(\frac{1}{36}\right)$ but a total of 7 in 6 ways $\left(\frac{6}{36}\right)$. So the students are not equally likely to be chosen. (Student 1 also has no chance.)

(b) Throw the die once: if the result is odd the chosen student comes from 1–6, if even from 7–12. Throw again: the score s gives student s (odd first throw) or $s+6$ (even first throw). Each of the 12 outcomes has probability $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$